

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (EE)

November 2013

EE 401 Electrical Machine Design I

Date: 12.11.2013, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Briefly explain the principles of core design of a current transformer. [04]

OR

(a) List the different types of duties for electrical machines. Draw the duty cycle for any one duty. [04]

Q - 1 (b) Answer the following questions with respect to transformer design. [03]
(Be brief and to the point).

(1) Explain the reason for using stepped core construction.

(2) If a designer selects higher value of flux density what will be its effect on performance and cost of the transformer?

(3) Why tapings are usually provided on the h.v. winding side?

Q - 2 Attempt any **TWO**. [14]

(a) What is design optimization? Derive necessary condition for designing a transformer with minimum cost and losses.

(b) Determine the main core dimensions for a 250 KVA, 6600/500V, 50 Hz, 3-phase star/delta core type transformer from the following data: Window space factor = 0.27, Current density = 2.5 A/mm², Max. flux density = 1.25 Wb/m², Volts per turn = 8.5 V, Use 4-stepped core limb section which has the area factor = 0.62
Height of window / width of window = 2

(c) A three phase 50 Hz oil cooled core type transformer has the following dimensions: distance between core centers 0.2 m, height of limb 0.24 m, diameter of first turn around limb 0.14 m, the flux density in the core is 1.25 tesla and current density in the conductors is 2.5 A/mm². Estimate the kVA rating. Assume a window space factor of 0.2 and the core is 2 stepped.

Q - 3 Attempt any **TWO**. [14]

(a) Estimate the p.u. regulation of the transformer referred to h.v. side, at full load and 0.8 pf lagging, 750 kVA, 6600V, 50 Hz, 3 phase, delta/star, and core type transformer has the following data: width of l.v. winding 30 mm, width of h.v. winding 25 mm, width of duct between h.v and l.v winding 15mm, length of the core 0.4 m, length of mean turn 1.5 m, h.v. winding turns 217, if the resistance per phase referred to the h.v side is 0.8 ohm.

- (b) Estimate the number of cooling tubes for a 250 KVA, 6600/400 V, 50 Hz, 3-phase delta / star core type oil cooled transformer from the following data: Temp rise = 50°C , Total losses at 90°C are 5.0 KW, Tank size = $125 \times 100 \times 50 \text{ cm}$. (h x l x w), Oil level = 115 cm., Sp-heat dissipation of plain tank = $12.5 \text{ W/m}^2/^{\circ}\text{C}$, Diameter of cooling tube = 5.0 cm. Show the tube arrangement by sketch.
- (c) For a 125 kVA, 2000/400 V, 50 Hz, single phase, shell type, oil immersed, self cooled transformer calculate (i) number of turns in lv and hv windings (ii) conductor sections of lv and hv winding. Assume suitable data for constants and other necessary factors.

SECTION – II

- Q - 4 5 kW, 250 V, 4 pole, 1500 rpm, shunt generator has square pole face. Average flux density in gap = 0.42 Tesla, ampere conductors/metre = 15000, find the main dimensions of the machine. Full load efficiency is 87% & ratio of pole arc to pole pitch = 0.66. [07]
- Q - 5 (a) Derive the output equation of a D.C. machine and explain its significance. [07]
- (b) Determine the main dimensions, number of poles and the length of air gap of a 600kW, 500V, 900 rpm generator. Assume $B_{av} = 0.6 \text{ wb/m}^2$, $a_c = 35000$ per meter, $\Psi = 0.75$ and efficiency = 91%. Now the design constraints are: Peripheral speed should not be more than 40 m/s, frequency of flux reversal should not be more than 50 Hz, current per brush arm should not be more than 400A and armature mmf per pole should not be more than 7500A. The mmf required for the air gap is 50 % of armature mmf and $kg = 1.15$. [07]

OR

- (b) A shunt field coil has to develop an mmf of 9000A. The voltage drop in the coil is 40V and resistivity of round wire is $0.021 \Omega/\text{m}$ and mm^2 . The depth of winding is 35 mm approximately and the length of mean turn is 1.4 m. Design a coil so that the power dissipated is 700 W/m^2 of the total coil surface. Take the diameter of the insulated wire 0.2 mm greater than that of bare wire. [07]
- Q - 6 Attempt any TWO. [14]
- (a) Design & develop a mush winding for a 4 pole, 24 slots, 3 phase stator of AC machine.
- (b) Find the front pitch, back pitch, winding pitch and commutator pitch for a simplex wave wound 13 slots, 4 pole dc armature with 13 commutator segments. Draw the winding diagram in developed form. Also draw sequence diagram to indicate the position of brushes. Assume number of coil sides per slot = 2.
- (c) A 4 pole generator supplies a current of 140A. It has 480 armature conductors wave connected. The brushes are given an actual lead of 10° . Calculate the cross and demagnetizing mmf per pole in each case. The field winding is shunt connected and takes a current of 10A; find the n number of extra shunt field turns to neutralize the demagnetization.
